Critical Slowing Down

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Abstract

This project explores critical slowing down in the two-dimensional Ising model near the critical temperature T_c. Using Markov chain Monte Carlo methods, we implement both the Metropolis and Wolff algorithms to generate spin configurations and measure autocorrelation times of magnetization and energy. Magnetic susceptibility is determined with an external field B, and, in the absence of the field, via the fluctuation-dissipation theorem (FDT). The scaling of relaxation times allows estimation of the dynamic critical exponent z for each algorithm. Comparisons between local and cluster updates highlight their relative efficiency at criticality, showing how algorithmic choices affect the sampling of Markov chains in systems exhibiting critical slowing down and field-induced responses.

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1 Introduction

The Ising model is a fundamental model in statistical mechanics used to study phase transitions and critical phenomena. Despite its seemingly simple definition, including binary spin variables and nearest-neighbor interactions, it captures the essential features of spontaneous symmetry breaking and critical behaviour observed in even more complex systems. In two dimensions, the Ising model exhibits a well-known second-order phase transition at a critical temperature T_c, where various physical quantities such as magnetization, susceptibility, and correlation length display power-law divergences characterized by critical exponents.

To study both static and dynamic properties of the Ising model, Monte Carlo simulations based on Markov-chain sampling are widely used. Near the critical point, these algorithms are subject to **critical slowing down**, where the relaxation and autocorrelation time increase rapidly as the correlation length diverges. In practice, this means that successive configurations generated by the simulation become highly correlated, leading to inefficient sampling and large statistical errors. Understanding this is essential for accurate numerical studies close to T_c.

This study investigates the effective dynamics of the 2D Ising model as generated by different Monte Carlo update algorithms: the local Metropolis algorithm, which performs local single-spin updates, and the global Wolff cluster algorithm, which employs non-local cluster flips. Although both algorithms sample from the same equilibrium distribution, they impose different dynamical behaviors on the system, leading to distinct scaling behaviours of the autocorrelation time near the critical point. This scaling is characterized by the **dynamic critical exponent** z, which determines the divergence of time scales according to $\tau \sim \xi^z$ as the correlation length $\xi \to \infty$ at criticality.

The goals of this report are:

1. To derive and discuss the theoretical background of dynamic critical behaviour and the definition of the dynamic exponent z.
2. To measure autocorrelation times of magnetization and energy for both Metropolis and Wolff dynamics as the temperature approaches T_c.
3. To compare the scaling and efficiency of the two algorithms in the critical region.

Section 2 introduces the theoretical framework, including the Ising model, Monte Carlo dynamics, and the concept of dynamic scaling. Section 3 outlines the simulation methodology and measurement procedures. Section 4 presents the numerical results and analysis, while Section 5 discusses the implications of the findings and compares algorithmic performance. Finally, Section 6 summarizes the main conclusions.

2 Theoretical Background

2.1 The 2D Ising Model

The two-dimensional square-lattice Ising model is a paradigmatic system for studying phase transitions and critical phenomena. Spins $s_i = \pm 1$ occupy the sites of the $L \times L$ lattice with periodic boundary conditions (PBD) and interact with their nearest neighbors. The Hamiltonian for the system is given by:

$$H = -J \sum_{\langle i,j \rangle} s_i s_j - h \sum_i s_i, \tag{1}$$

where J is the interaction strength, h is an external magnetic field, and the sum $\langle i, j \rangle$ runs over nearest neighbor pairs.

In zero external field ($h = 0$), the model exhibits a second-order phase transition at a critical temperature T_c, separating a low-temperature ferromagnetic phase (with non-zero spontaneous magnetization) from a high-temperature paramagnetic phase (with zero magnetization). For the 2D Ising model, the critical temperature is known exactly from Onsager's solution:

$$\frac{k_B T_c}{J} = \frac{2}{\ln(1 + \sqrt{2})} \approx 2.2691853, \tag{2}$$

or equivalently,

$$J\beta_c = J \frac{1}{k_B T_c} \approx 0.4406868, \tag{3}$$

where k_B is the Boltzmann constant. [1]

The order parameter is the magnetization per spin, defined as:

$$m = \frac{1}{N} \sum_{i=1}^{N} s_i, \tag{4}$$

where $N = L^2$ is the total number of spins in the lattice.

The zero-field susceptibility χ is defined thermodynamically as:

$$\chi = \frac{\partial \langle m \rangle}{\partial h}\bigg|_{h=0}. \tag{5}$$

By the fluctuation-dissipation theorem (FDT), this equals to

$$\chi = \beta N (\langle m^2 \rangle - \langle m \rangle^2). \tag{6}$$

In finite simulations at $h = 0$, the $\mathbb{Z}_2$ symmetry implies $\langle m \rangle = 0$ unless symmetry is explicitly broken. For thermodynamic estimates we use m in the fluctuation formula for χ, but for visualization we plot $\langle |m| \rangle$ in figures (in particular for the Wolff algorithm), since $|m|$ avoids sign cancellations and makes the ordered phase evident.

In practice, this derivative is often approximated by a finite-difference scheme using a small nonzero field h_0:

$$\chi \approx \frac{\langle M \rangle_{+h_0} - \langle M \rangle_{-h_0}}{2h_0}. \tag{7}$$

2.2 Monte Carlo Dynamics

2.2.1 Markov Chains and Master Equations

Monte Carlo updates define a Markov chain on the configuration space with a transition operator that has the Boltzmann distribution as its stationary measure. For relaxational models, such as the stochastic Ising model, the time-dependent behavior is described by a master equation ([2])

$$\frac{\partial P_n(t)}{\partial t} = - \sum_{n \neq m} \left[P_n(t) W_{n \to m} - P_m(t) W_{m \to n} \right], \tag{8}$$

where $P_n(t)$ is the probability of the system being in state n at time t, and $W_{n\to m}$ is the transition rate for $n \to m$.

The solution to the master equation is a sequence of states, but the time variable is a stochastic quantity which does not represent true time. A relaxation function $\phi(t)$ can be defined which describes time correlations *within equilibrium*

$$\phi_{MM}(t) = \frac{\langle M(0)M(t)\rangle - \langle M\rangle^2}{\langle M^2\rangle - \langle M\rangle^2}. \tag{9}$$

When normalized in this way, the relaxation function is 1 at $t = 0$ and decays to zero as $t \to \infty$. It is important to remember that for a system in equilibrium any time in the sequence of states may be chosen as the '$t = 0$' state. The asymptotic, long time behavior of the relaxation function is exponential, i.e.

$$\phi(t) \to e^{-t/\tau}, \tag{10}$$

where the correlation time τ diverges as T_c is approached.

2.2.2 Dynamic Critical Phenomena

This dynamic (relaxational) critical behavior can be expressed in terms of a power law as well,

$$\tau \propto \xi^z \propto \varepsilon^{-\nu z}, \tag{11}$$

where ξ is the (divergent) correlation length, $\varepsilon = |1 - T/T_c|$, and z is the dynamic critical exponent.

In a finite system of linear size L, the divergence of the correlation length at criticality is limited by the system size. Exactly at T_c, the correlation length therefore saturates at $\xi \sim L$. Inserting this finite-size cutoff into the critical slowing-down relation $\tau \propto \xi^z$ gives

$$\tau_c \sim L^z. \tag{12}$$

Thus, at the critical temperature, the relaxation time is no longer controlled by the reduced temperature ε, but instead by the system size itself. As L increases, correlations require more time to propagate across the entire system, leading to a relaxation time that scales as a power law in L with exponent z.

A second relaxation time, the integrated relaxation time, is defined by the integral of the relaxation function

$$\tau_{\text{int}} = \int_0^\infty \phi(t)\, dt. \tag{13}$$

This quantity is of particular importance for the determination of errors and is expected to diverge with the same dynamic exponent as the 'exponential' relaxation time, as both are governed by the same long-time behavior of $\phi(t)$.

3 Methodology

3.1 Model and Setup

We simulate the ferromagnetic Ising model on a two-dimensional square lattice of linear size L with PBC. The number of spins is $N = L^2$, with spin variables $s_i \in -1, +1$. Throughout we set $k_B = 1$, hence $\beta = \frac{1}{T}$, and $J = 1$.

Lattice and sizes:

- $L \in [2, 4, 8, 16, 32, 64, 128]$
- PBC in both directions.

Temperature range:

- We scan a window around the exact critical point $\beta_c = 0.4407$, namely

  $$\beta \in [0.1, 0.2, 0.3, 0.4, 0.41, 0.42, 0.43, 0.44, 0.4407, 0.45, 0.46, 0.47, 0.48, 0.49, 0.5, 0.6].$$

  For the Wolff algorithm, we also included $\beta = 1$.
- For finite-size scaling at T_c, we fix $\beta = \beta_c$ and vary L.

Initialization:

- Hot starts, i. e. spins initialized independently to ± 1 with equal probability.
- Burn-in was excluded via visual examination in time series of energy and magnetization.

Time units:

- Metropolis: one Monte Carlo sweep (MCS) = N single-spin flip attempts.
- Wolff: one Wolff step = one cluster flip; to compare time scales, we define 'Wolff sweeps' such that on average N spins are flipped in one sweep, as detailed in section 3.2.2.

Observables:

- Energy E, magnetization M.
- For dynamic analysis: time series of M and E to compute $\rho_A(t)$, τ_{int}, and τ_{exp}.

3.2 Update Algorithms

3.2.1 Metropolis (local)

Metropolis method [3] configurations are generated from a previous state using a transition probability which depends on the energy difference between the initial and final states. The states produced follows a time ordered path, and this time is not a proper time but a Monte-Carlo time. For relaxational models, such as the stochastic Ising model, the time-dependent behavior is described by the master equation 8, with the probability of the nth state occurring in a classical system is given by $P_n = \frac{e^{-\beta E_n}}{Z}$ with the partition function Z which we don't know. To bypass the expression of Z, we generate a Markov chain of states, i.e. we generate each new state directly from the preceding state. If we produce the n_{th} state from the m_{th} state, and then calculate the relative probability which is the ratio of the individual probabilities, the denominator cancels. As a result, only the energy difference between the two states is needed: $\Delta E = E_n - E_m$.

Now thanks to the master equation in equilibrium $P_n W_{n\to m} = P_m W_{m\to n}$:

$$\frac{Wn \to m}{Wm \to n} = \frac{P_m}{P_n} = e^{-\beta \Delta E}. \tag{14}$$

Thus

$$P_{n\to m} = \begin{cases} e^{-\beta\Delta E} & \text{for } \Delta E > 0 \\ 1 & \text{for } \Delta E \leq 0 \end{cases} \tag{15}$$

The steps of the algorithm are as follows [4]:

1. Choose an initial state.
2. Choose a lattice site i randomly.
3. Compute the energy change ΔE resulting from flipping the spin at site i.
4. Generate a random number $0 < r < 1$.
5. If $r < e^{-\beta \Delta E}$, flip the spin.
6. Move to the next site and return to step (3).

3.2.2 Wolff (cluster)

Wolff ([5]) proposed an alternative algorithm based on the Fortuin-Kasteleyn theorem in which single clusters are grown and flipped sequentially. The algorithm begins with the random choice of a single site. Bonds are then drawn to all nearest neighbors which are in the same state with probability

$$p = 1 - e^{-K\delta_{\sigma_i \sigma_j}}. \tag{16}$$

One then moves to all sites which have been connected to the initial site and places bonds between them and any of their nearest neighbors which are in the same state with probability given by equation 16. The process continues until no new bonds are formed and the entire cluster of connected sites is then flipped. Another initial site is chosen and the process is then repeated. A single Wolff cluster update consists of the following steps [2]:

1. Randomly select a seed spin from the lattice.
2. Grow a cluster by adding neighboring spins with the same orientation with probability $p_{\text{add}} = 1 - \exp(-2\beta J)$, where J is the nearest-neighbor coupling and $\beta = 1/T$ is the inverse temperature.
3. Repeat the cluster growth iteratively for all new cluster members until no further spins are added.
4. Flip all spins in the cluster simultaneously.

Depth-first search is used to traverse and grow the cluster efficiently. DFS avoids revisiting the same spins multiple times, making the algorithm fast even for very large clusters.

The Monte Carlo time was rescaled for the Wolff algorithm using the mean number of flipped clusters $\langle C \rangle$ per step [2]:

$$\Delta t = \frac{\langle C \rangle}{L^2}, \quad t_{\text{MC}} = \sum \Delta t \tag{17}$$

This makes the Wolff steps comparable to a full Metropolis sweep. Time series of energy E, magnetization M, absolute magnetization $|M|$, and squared magnetization M^2 were analyzed for each β to study fluctuations and equilibrium behavior.

3.3 Practical Estimators and Extraction of z

To quantify critical slowing down we compute the autocorrelation function of an observable A (for instance the magnetization M, the energy or the susceptibility):

$$C_A(t) = \langle A_s A_{s+t} \rangle - \langle A \rangle^2, \tag{18}$$

and the normalized autocorrelation (autocorrelation coefficient)

$$\rho_A(t) = \frac{C_A(t)}{C_A(0)}. \tag{19}$$

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A convenient scalar measure of the memory of the Markov chain is the *integrated autocorrelation time*

$$\tau_{\text{int},A} = \frac{1}{2} + \sum_{t=1}^{T_{\max}} \rho_A(t), \tag{20}$$

where the sum is truncated at a cutoff $T_{\max}$. A common choice of $T_{\max}$ is a multiple of $\tau_{\mathrm{int},A}$ itself, i.e. $T_{\max} = c\tau_{\mathrm{int},A}$, where c is chosen to be 5. Since the integrated autocorrelation is defined self-consistently, this is done in an iterative process.

We estimate the relaxation time of an observable A $\tau_A(L)$ by the integrated autocorrelation time and then fit $\log \tau_c$ versus $\log L$ to extract z:

$$\log \tau_c = z \log L + \text{const.} \tag{21}$$

4 Results

4.1 Equilibrium Verification

We verified equilibrium behavior by measuring $\langle E(T)\rangle$ and $\langle m(T)\rangle$ at our largest lattice size $L = 128$ after the burn-in. The curves reproduce the expected ferromagnetic ordering at low T and disordered phase at high T, with susceptibility peaks near T_c. This validates our simulation settings used for the dynamical analysis.

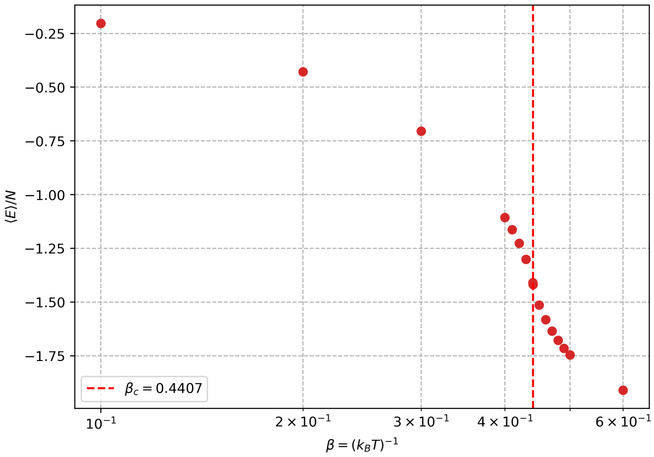

Figure 1: Energy per spin versus $\frac{\langle E\rangle}{N}$ inverse temperature β for the 2D Ising model using Metropolis updates. Points show the equilibrium average energy.

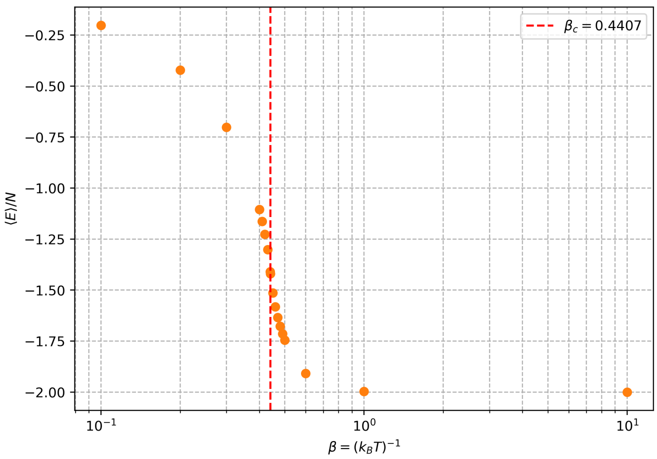

Figure 2: Energy per spin versus $\frac{\langle E\rangle}{N}$ inverse temperature β for the 2D Ising model using Wolff updates. Points show the equilibrium average energy.

In figures 1 and 2 we can see that for both algorithms, as β increases, $\frac{\langle E\rangle}{N}$ decreases smoothly and exhibits a pronounced change in slope near the critical $\beta_c \approx 0.4407$ (vertical reference line).

Magnetization plots in figures 3 and 4 show the expected spontaneous magnetization below T_c and decay to zero above T_c for both algorithms, confirming correct equilibrium sampling.

The peak of the susceptibility in figure 5 near β_c reflects the divergence of χ at the phase transition, which proves the second-order phase transitions. Before β_c (high temperatures, $T > T_c$), χ the system is in the paramagnetic phase, spins are disordered and fluctuations are small, after β the system is in the ferromagnetic phase, most spins are aligned, so fluctuations of the total magnetization decrease.

The maximum susceptibility is very high, which is typical for finite-size systems simulated with

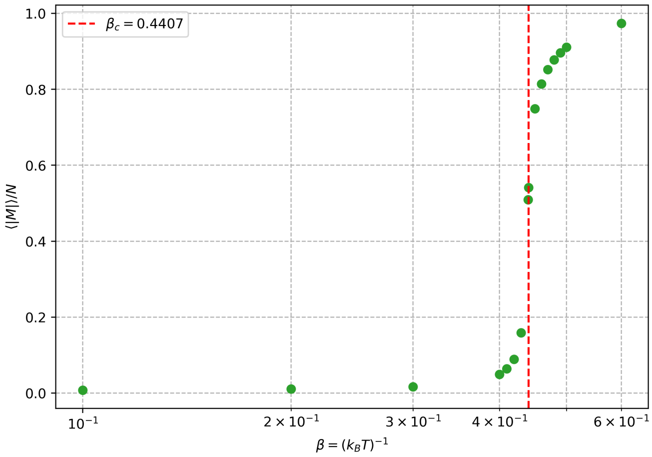

Figure 3: Absolute magnetization versus inverse temperature β using Metropolis updates. Points show the equilibrium average magnetization. The red dashed line indicates the critical value β_c.

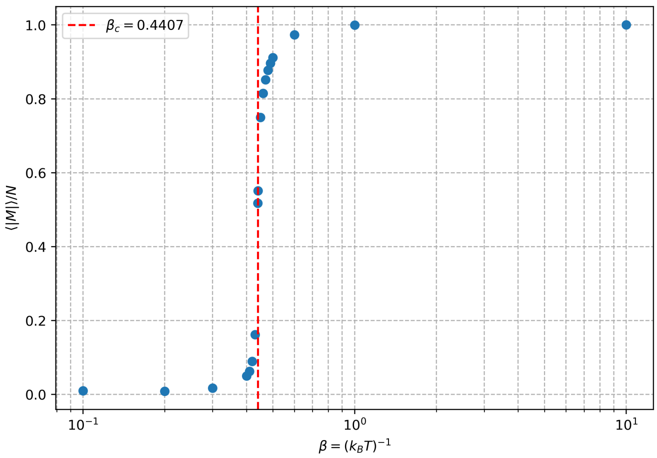

Figure 4: Magnetization versus inverse temperature β using Wolff updates. Points show the equilibrium average magnetization. The red dashed line indicates the critical value β_c.

Metropolis: the peak would diverge in the thermodynamic limit. Small fluctuations for low β (high T) are expected because the system behaves almost like independent spins. The asymmetry of the peak is also visible: the high-β side (low T) decays more slowly than the low-β side, consistent with finite-size effects and critical slowing down in Metropolis simulations.

Then we can compare the value of the susceptibility with the FDT and with a small non zero external field.

The susceptibility obtained from the fluctuation–dissipation theorem (FDT) and the one computed using a small external field are different near the critical point. In our plot, the field $h_0 = 0.002$ turns out to be too large near the critical point: it suppresses the true divergence of χ, producing the plateau observed for $\beta > \beta_c$. This cutoff effect is enhanced by finite-size limitations, which also prevent the susceptibility from diverging. As a result, both estimators are consistent away from criticality, but the finite-field approach underestimates the critical behaviour unless h is reduced by several orders of magnitude. However, using a small external field provides a more controlled and numerically stable way to measure the susceptibility. It fixes the direction of the magnetization, avoids symmetry-related cancellations, and gives a direct estimate of the response $\partial M/\partial h$.

Near the critical temperature, the magnetic susceptibility χ is highly sensitive to the efficiency of the Monte Carlo algorithm. In simulations using the Metropolis method, which updates one spin at a time, large correlated clusters evolve very slowly, causing successive configurations to remain

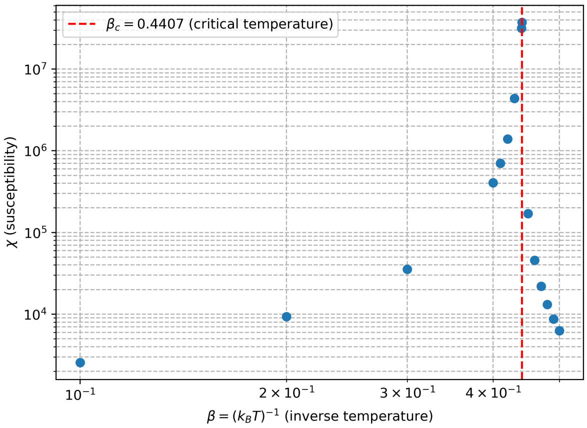

Figure 5: Susceptibility versus inverse temperature β using Metropolis updates. Points show the equilibrium susceptibility computed via fluctuation-dissipation theorem. The red dashed line indicates the critical value β_c.

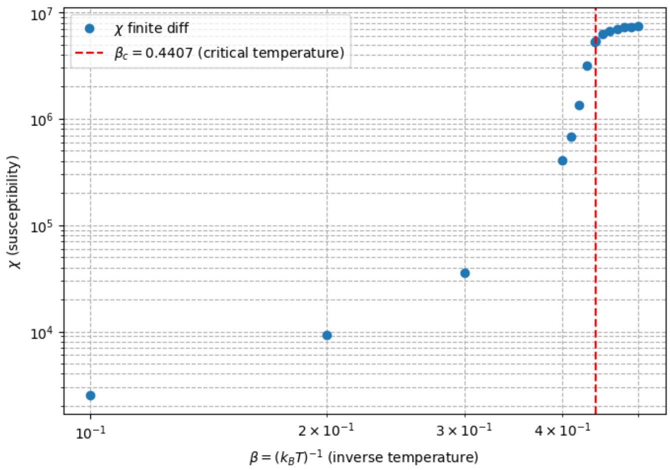

Figure 6: Susceptibility as a function of β with small external field. The red dashed line indicates the critical value β_c.

strongly correlated. This results in an artificial amplification of magnetization fluctuations, with the peak susceptibility reaching exceedingly high values on the order of 10^7. By contrast, the Wolff algorithm flips entire clusters of spins in a single update, efficiently breaking long-range correlations and producing largely independent configurations. Consequently, the susceptibility obtained with Wolff is significantly lower, with its peak around 10^{-2}.

It is to be noted that for the Wolff algorithm, we plotted the susceptibility computed from the absolute magnetization $|M|$ rather than M itself, which prevents sector cancellations, as due to $\mathbb{Z}_2$ symmetry finite systems flip between magnetized sectors. This yields the standard finite-size peak near β_c. That choice does not affect our determination of T_c or the scaling analysis.

4.2 Autocorrelation Behavior

We illustrate the dynamical contrast between local and cluster updates in figure 8 by plotting the integrated autocorrelation time of the energy versus β at fixed $L = 128$.

For energy, Metropolis shows a pronounced peak centered near $\beta_c \approx 0.4407$ at $\tau_{int} = 206.97$, evidencing strong critical slowing down and long relaxation at the transition. By contrast, Wolff yields a small $\tau_{int} = 2.94$ at β_c, demonstrating that cluster updates suppress critical correlations and greatly reduce relaxation times. Thus, the Metropolis relaxation is slower by a factor of ≈ 70.

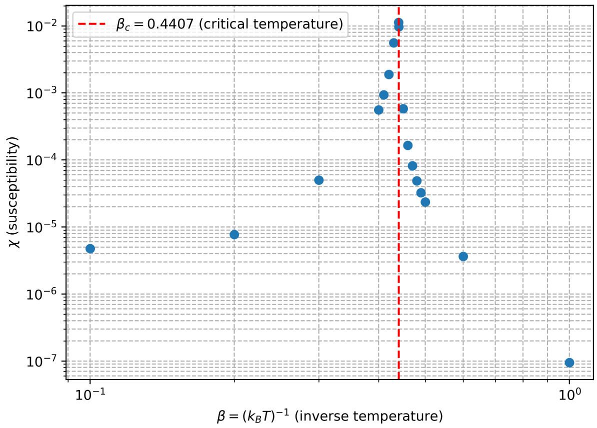

Figure 7: Susceptibility versus inverse temperature β using Wolff updates. Points show the equilibrium susceptibility computed via fluctuation-dissipation theorem. The red dashed line indicates the critical value β_c.

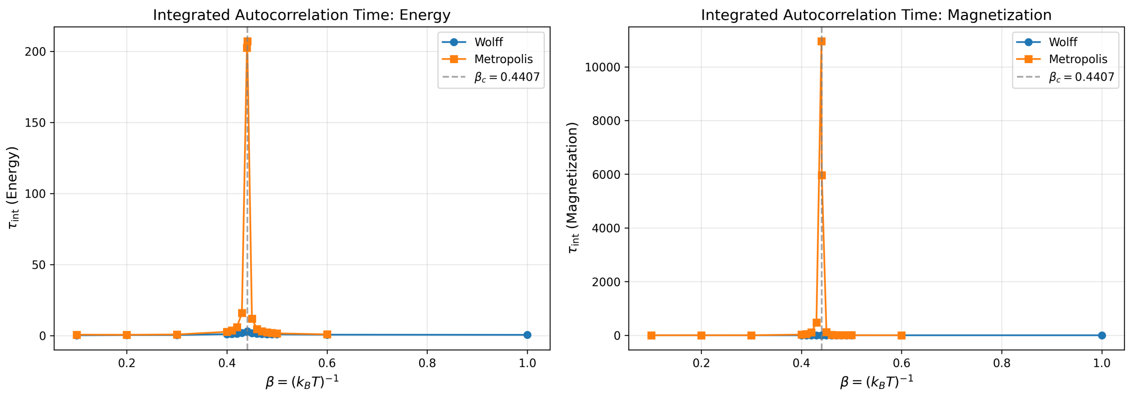

Figure 8: Integrated autocorrelation functions $\tau_{int,M}(t)$ of the magnetization for $L = 128$ at criticality $\beta_c \approx 0.4407$, comparing Metropolis (blue) and Wolff (orange) updates.

For magnetization it is qualitatively the same. Metropolis shows a clear peak centered near $\beta_c \approx 0.4407$ at $\tau_{int} = 10948.72$. Conversely, Wolff yields $\tau_{int} = 1.59$ at β_c. Hence, the Metropolis relaxation is slower by a factor of $\approx 6.9 \times 10^3$.

This qualitative difference motivates our finite-size scaling of τ at β_c to extract the dynamic exponent z.

4.3 Finite-Size Scaling and Extraction of z

For each lattice size L, we load the Monte Carlo time series of magnetization and energy at the critical inverse temperature $\beta_c = 0.4407$. We compute the normalized autocorrelation function using the fast Fourier transform (FFT), and determine the integrated autocorrelation time τ_{int} using Sokal's self-consistent method as described above (3.3). An iterative scheme is applied to automatically estimate and remove the thermalization (burn-in) period while ensuring that the remaining data satisfy $N \gg 50\,\tau_{\mathrm{int}}$. The resulting autocorrelation times for each L are then fitted to the expected dynamic scaling form $\tau_{\mathrm{int}}(L) \sim L^z$ in log-log scale. In this scale, we can fit a straight line to the data points with the slope representing the dynamic critical exponent z. This allows us to extract the dynamic critical exponent z for both the magnetization and the energy via the slope. These plots can be seen in figures 9 and 10.

The extracted dynamic exponents show a stark algorithmic contrast. For Metropolis we find $z_M \approx 2.03 \pm 0.03$ (magnetization) and $z_E \approx 1.39 \pm 0.06$ (energy). The magnetization result is close

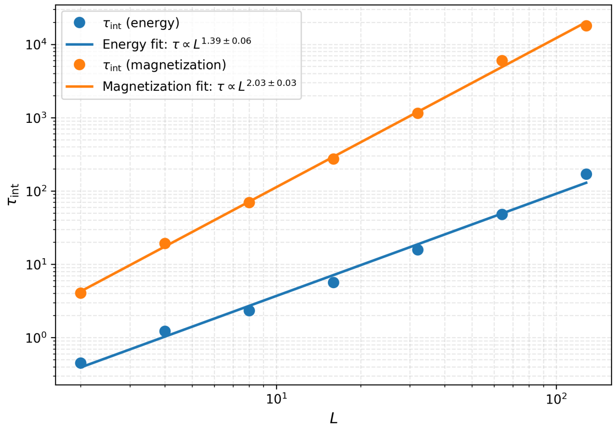

Figure 9: Finite-size scaling of integrated autocorrelation time τ versus system size L at criticality $\beta_c \approx 0.4407$ for energy (blue) and magnetization (orange) using Metropolis updates.

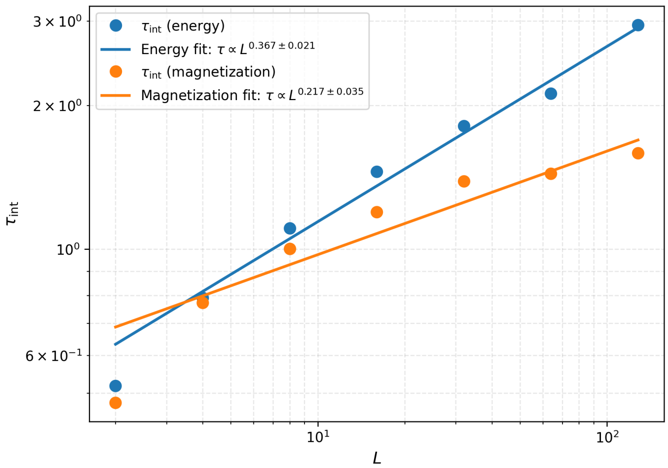

Figure 10: Finite-size scaling of integrated autocorrelation time τ versus system size L at criticality $\beta_c \approx 0.4407$ for energy (blue) and magnetization (orange) using Wolff updates.

to the expected $z \approx 2.1$ ([6]) for local dynamics, while the energy-based estimate is noticeably lower, reflecting that τ_E is harder to measure reliably, because energy time series have smaller signal-to-noise, whereas magnetization directly probes the slow order-parameter mode that controls critical relaxation. Others [7] have also found differences in the relaxation behavior of "energy-like" and "magnetization-like" observables. For Wolff, $z_M \approx 0.217 \pm 0.035$ and $z_E \approx 0.367 \pm 0.021$ both indicate a dramatic suppression of critical slowing down. The magnetization-based value lies within the expected value of $z \approx 0.29$ [6] obtained by using the integrated autocorrelation time. Other studies find a $z \approx 0$ by directly measuring corrleation length [8] highliting that different methods can lead to different results. The energy-based value is somewhat higher, again consistent with noisier τ_E estimates and the fact that cluster flips most efficiently decorrelate the order parameter. Overall, τ_M provides the more reliable estimator of z for both algorithms, with τ_E tending to bias low for Metropolis and high for Wolff due to noise and mixed time scales.

4.4 Comparison of Algorithm Efficiency

The Wolff algorithm drastically reduces critical slowing down by updating large groups of spins simultaneously rather than flipping spins individually as in local-update methods like Metropolis. Near the critical temperature, the system develops large clusters of correlated spins, which evolve very slowly under single-spin updates, leading to long autocorrelation times and artificially large fluctuations in observables such as magnetization. This is evidenced in Metropolis simulations, where the integrated autocorrelation time for energy peaks at $\tau_{\text{int}} \approx 207$ and for magnetization at

$\tau_{\text{int}} \approx 10{,}949$ near β_c, indicating very slow relaxation. Consequently, the susceptibility is also overestimated, reaching values as high as 10^7. In contrast, cluster updates flip entire clusters of spins in a single move, effectively breaking long-range correlations and producing largely independent successive configurations. This results in drastically reduced autocorrelation times—$\tau_{\text{int}} \approx 2.94$ for energy and $\tau_{\text{int}} \approx 1.59$ for magnetization at β_c—and a realistic estimation of susceptibility with a peak around 10^{-2}. By efficiently decorrelating the system, cluster algorithms suppress critical slowing down and allow for faster convergence near the phase transition.

5 Conclusion

- Summarize key findings: - Divergence of τ near T_c,

- Distinct dynamic exponents for different algorithms,

- Algorithmic efficiency and importance for Monte Carlo near criticality.

- Mention physical significance: critical slowing down as an emergent dynamical effect in critical systems.

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